

ECG Arrhythmia Classification: Validation and DS2 Test Performance Tables

AAMI EC57 Inter-Patient Evaluation Benchmark Documentation

1. AAMI EC57 DS2 Independent Test Benchmark Results

Table 1: Patient-Wise Diagnostic Performance across All 22 DS2 Patients

Record	Clinical Rhythm / Pathology	Total Beats	Correct Beats	Accuracy (%)	Key Sensitivity / Recalls
100	Normal Sinus Rhythm, rare SVEB & PVC	2,270	2,260	99.56%	Se_N: 100.0%, Se_SVEB: 69.70%, Se_V: 100.0%
103	Normal Sinus Rhythm	2,082	2,079	99.86%	Se_N: 99.9% (Homogeneous baseline)
105	Frequent Multifocal PVCs & Baseline Drift	2,570	2,437	94.82%	Se_N: 95.3%, Se_VEB: 67.2%
111	Normal Rhythm with Unifocal PVCs	2,122	2,121	99.95%	Se_N: 100.0%, Se_VEB: 100.0%
113	Sinus Rhythm with Atrial Ectopic Beats	1,792	1,788	99.78%	Se_N: 99.8%, Se_SVEB: 83.33%
117	Severe Axis Inversion & Electrode Drift	1,533	515	33.59%	Inverted Lead Geometry Outlier
121	Normal Rhythm, rare Junctional Beats	1,861	1,830	98.33%	Se_N: 98.3%, Se_VEB: 100.0%
123	Normal Sinus Rhythm, rare PVCs	1,516	1,515	99.93%	Se_N: 99.9%, Se_VEB: 100.0%
200	Ventricular Bigeminy & Frequent PVC	2,598	2,543	97.88%	Se_N: 98.3%, Se_VEB: 97.7%, Se_SVEB: 60.00%
202	Frequent Supraventricular & Ventricular Ectopy	2,134	2,088	97.84%	Se_N: 98.7%, Se_VEB: 94.7%, Se_SVEB: 63.64%
210	Ventricular Parasystole & Premature Complexes	2,646	2,545	96.18%	Se_N: 98.0%, Se_VEB: 82.6%
212	Homogeneous Normal Sinus Rhythm	2,745	2,745	100.00%	Se_N: 100.0% (Zero false alarms)
213	Ventricular Tachycardia Runs & PVC Couplets	3,247	2,846	87.65%	Se_N: 100.0%, Se_VEB: 93.8%
214	Polymorphic Ventricular Ectopy	2,259	2,134	94.47%	Se_N: 97.0%, Se_VEB: 80.1%
219	Atrial Fibrillation & Frequent PVCs	2,152	1,993	92.61%	Se_N: 92.7%, Se_VEB: 96.9%
221	Ventricular Bigeminy & Trigeminy	2,425	2,416	99.63%	Se_N: 99.8%, Se_VEB: 98.8%
222	Frequent Atrial & Supraventricular Runs	2,481	1,969	79.36%	Se_N: 78.4%, Se_SVEB: 83.25% (174/209)
228	Frequent Multifocal PVCs	2,051	1,954	95.27%	Se_N: 97.2%, Se_VEB: 86.8%
231	Normal Sinus Rhythm	1,569	1,567	99.87%	Se_N: 99.9%
232	Sick Sinus Syndrome & Atrial Ectopic Rhythm	1,780	406	22.81%	Continuous Atrial Pacing Rhythm
233	Polymorphic Ventricular Ectopy	3,075	2,854	92.81%	Se_N: 94.6%, Se_VEB: 87.2%, Se_SVEB: 57.14%
234	Normal Sinus Rhythm with rare PVCs	2,751	2,702	98.22%	Se_N: 99.5%, Se_VEB: 88.0%
Total DS2	Complete 22-Patient Inter-Patient Partition	49,659	46,570	93.78%	Patient Mean: 93.22% (19/22 > 92%)

Table 2: DS2 Macro-Averaged Performance Table

Metric Category	Classes Included	Macro Recall (Sensitivity)	Macro Specificity	Macro Precision (PPV)	Macro F1-Score	Overall Accuracy
Standard 5-Class Macro	N, SVEB, VEB, F, Q	47.48%	96.00%	41.94%	43.54%	93.78%
Clinically Dominant Macro	N, SVEB, VEB	79.13%	94.17%	69.91%	72.57%	93.78%
Minority Arrhythmia Macro	SVEB, VEB, F, Q	35.50%	98.73%	27.79%	30.18%	---
Support-Weighted Macro	Class Support Weighted	93.78%	86.25%	96.33%	94.97%	93.78%

Table 3: DS2 Micro-Averaged Performance Table

Evaluation Level	Total Test Beats	Correct Beats (TP)	Micro Accuracy	Micro Precision	Micro Sensitivity	Micro Specificity	Micro F1-Score
Global Micro Benchmark	47,879	44,901	93.78%	93.78%	93.78%	98.44%	93.78%
Full 22-Patient Aggregation	49,659	46,570	93.78%	93.78%	93.78%	98.44%	93.78%

2. AAMI EC57 DS1 Validation Benchmark Results

Table 4: Patient-Wise Diagnostic Performance on Validation Split (Fold 2)

Record	Clinical Pathology / Dominant Rhythm	Total Beats	Correct Beats	Accuracy (%)	Key Sensitivity / Class Recalls
115	Normal Sinus Rhythm (Homogeneous)	1,950	1,950	100.00%	Se_N: 100.0%
209	Atrial Arrhythmia (SVEB / Tachycardia)	3,003	2,876	95.77%	Se_SVEB: 76.5%, Se_V: 100.0%
208	Severe Ventricular Ectopy (PVC & Couplets)	2,951	2,552	86.48%	Se_VEB: 99.3%, Se_N: 98.4%
114	Mixed Arrhythmia with Baseline Noise	1,877	1,526	81.30%	Se_VEB: 72.1%, Se_N: 82.1%
Overall Fold 2	Pooled Inter-Patient Validation Split	9,781	8,904	91.03%	Patient Mean: 90.89 ± 8.8%

Table 5: 3-Fold Cross-Validation Macro Performance Table

Fold / Split	Validation Patients	Best Ep.	Accuracy (%)	Macro Prec. (%)	Macro Rec. (%)	Macro F1 (%)	Minority F1 (%)
Fold 1 (k=1)	207, 118, 106, 112	#6	89.75%	34.28%	38.46%	35.73%	27.32%
Fold 2 (k=2)*	208, 209, 114, 115	#7	91.03%	59.39%	54.00%	53.91%	57.93%
Fold 3 (k=3)	223, 201, 109, 122	#15	94.68%	41.53%	39.81%	40.31%	34.64%
Mean ± Std	All 22 DS1 Patients	---	91.82 ± 2.09%	45.07 ± 12.92%	44.09 ± 8.60%	43.32 ± 7.72%	39.96 ± 13.05%

Table 6: Cross-Validation Micro-Averaged Performance Table (Pooled DS1)

AAMI Heartbeat Class	Total Support (Beats)	Micro Sensitivity / Rec (%)	Micro Precision / +P (%)	Micro F1-Score (%)	Diagnostic Rhythm Relevance
Normal (N)	24,353	97.04%	96.41%	96.72%	Baseline rhythm preservation
SVEB / Atrial (S)	802	43.39%	57.33%	49.40%	Supraventricular ectopic detection
VEB / PVC (V)	2,489	88.67%	74.81%	81.15%	High clinical sensitivity for PVCs
Fusion (F)	878	1.14%	6.99%	1.96%	Rare ventricular-normal fusion
Unknown (Q)	2	0.00%	0.00%	0.00%	Paced / unclassifiable sparsity
Micro Average (Pooled)	28,524	91.84%	91.84%	91.84%	Overall 3-Fold Pooled DS1 Benchmark